

Vector Class Reference

The Vector class represents a vector (x,y,z) in 3D space. [More...](#)

```
#include <vector.h>
```

Detailed Description

For the sake of simplicity, **Vector** is implemented as a typedef of Qt's [QVector3D](#).

Example of use:

```
#include "point.h"
#include "vector.h"

// create two vectors
Vector u = Vector(0.0, 0.0, 0.0);
Vector v = Vector(1.0, 0.0, 0.0);

// get components
float x = u.x();
float y = u.y();
float z = u.z();

// set components
u.setX(2.0);
u.setY(4.0);
u.setZ(2.0);

// get length
float len = u.length();

// normalize (in place)
u.normalize();

// get normalized copy
v = u.normalized();

// common operations
Vector w;
w = u + v; // vector addition
w = u - v; // vector subtraction
w = 2.0*u; // scalar multiplication
float dot = dotProduct(u, v); // dot product
w = crossProduct(u, v); // cross product
```